

Introduction to ORCAD

(EBN 111 : Electricity and Electronics)

Pre-Practicum 1



By Dieter Barnard

EBN 111: Practical 1 Objectives



- ❑ Familiarize students with OrCAD
- ❑ Execute circuit simulation in OrCAD
- ❑ Complete Practicum Assessment
- Practical 1 will take place during the week of
27 Feb – 2 Mar
- Group allocations will be posted on the website

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OrCAD



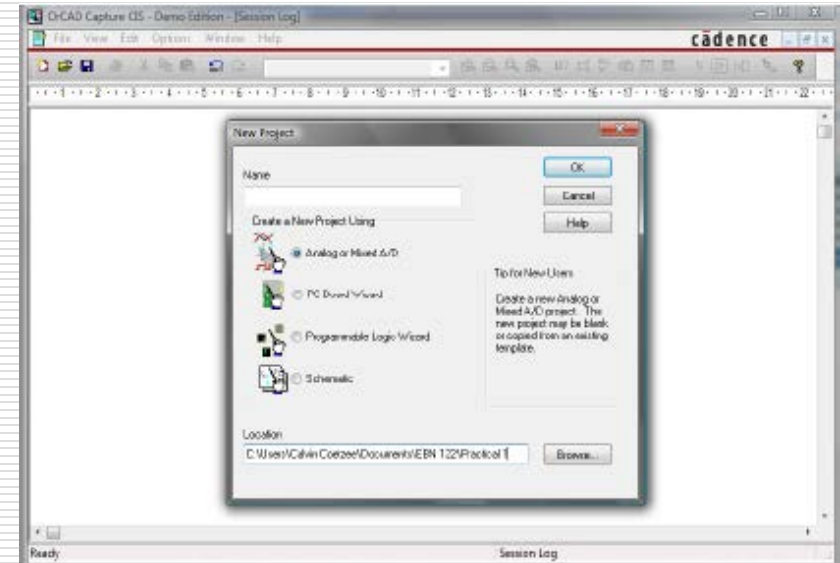
- ❑ Schematic Capture and circuit Simulation are two essential tools in computer aided circuit analysis and design
- ❑ Computer aided design makes it possible to deal with problems which are computationally very complex and lengthy without the aid of a computer
- ❑ Simulation software is part of modern engineering practice
- ❑ This practicum provides an opportunity to gain some skills in using these tools, which are used in many subsequent modules

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- ❑ What is Schematic Capture?
- ❑ Schematic Capture is an electronic means of graphically constructing a circuit in a format that can be subsequently read by software such as PSpice for circuit simulation and by OrCAD Layout for circuit board layout

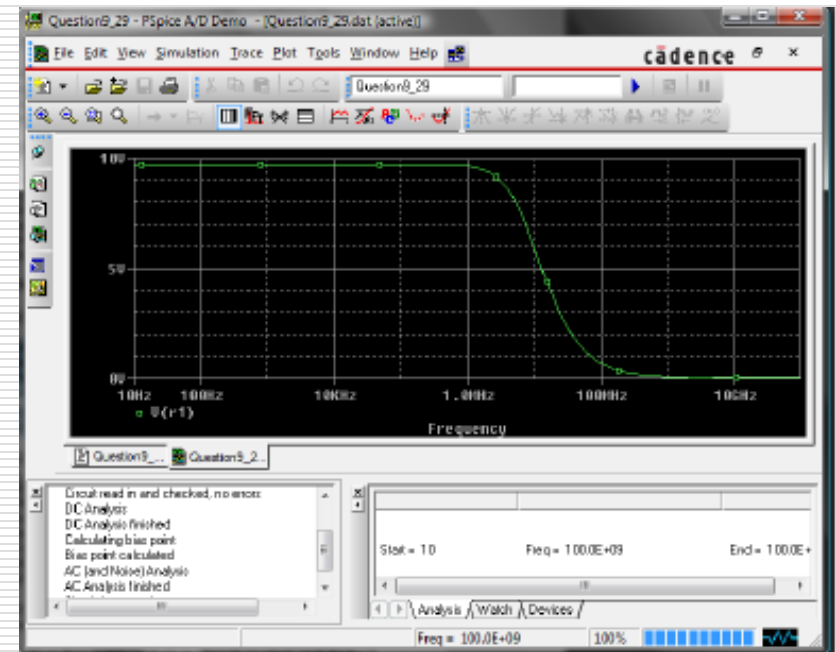


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- ❑ What is SPICE and what is PSpice?
- ❑ SPICE was originally an analogue circuit simulator software package that was developed at the University of California at Berkeley
- ❑ PSpice is a version of SPICE for personal computers

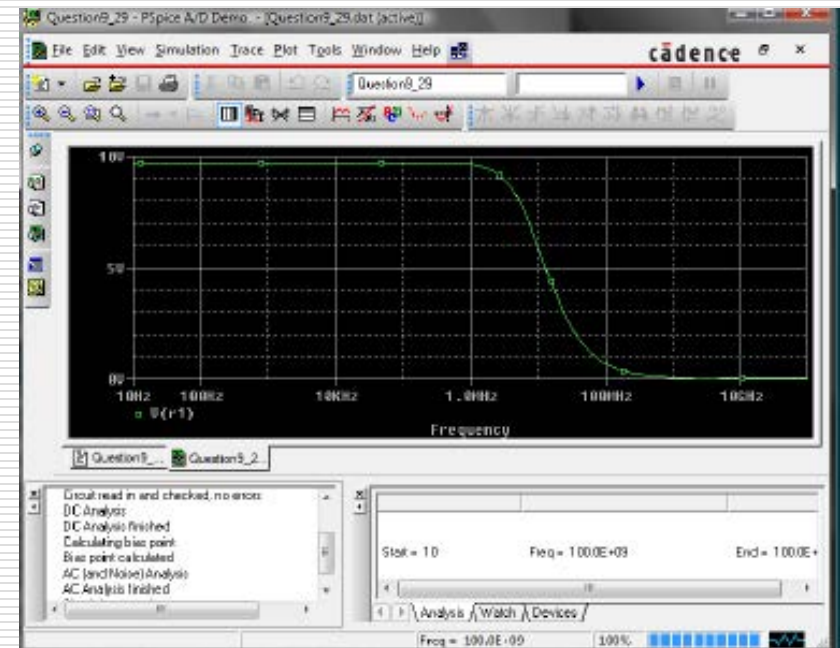


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- ❑ How does PSpice help in circuit analysis and design?
- ❑ PSpice's strong point is that it helps the user simulate and analyse the circuit design graphically on the computer without having to build a physical circuit



EBN 111: Practical 1 Preparation



- ❑ Log on to ClickUP and go to EBN111's page
- ❑ Click on the Practicals tab and select Practical 1
- ❑ Open the Practical 1 – Instructions.pdf file
- ❑ This file summarises what is required for practical 1.
- ❑ Perform all the tasks described in the Practical 1 – Instructions.pdf document

EBN 111: Practical 1

Practicum requirements



- What is required in order to complete the practical?
 - Complete the tutorial
 - Complete the assignment on ClickUP
 - Attend the 3 hour practical session

EBN 111: Practical 1 Tutorial



- ❑ In the Practical 1 – Instructions.pdf there is a link to a tutorial that helps you to download and familiarise yourself with the software

Complete this tutorial and then complete “Pre-practical Assignment” on ClickUP

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Pre-practical Assignment



- ☐ The Pre-practical assignment can be found on ClickUP and will be active from Friday 17 February – 24 February
- ☐ It must be completed during this week – doesn't have to be in 1 sitting – Make sure to save answers.
- ☐ After reading through the slides, completing the tutorial and submitting the pre-practical assignment, you should be sufficiently prepared for the practical session

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Demonstration of different simulations



- A demonstration is given in class:
 - how to complete different types of simulations, and
 - how to view and analyse the results
- You can download a student version of the software for working at home
- The OrCAD software is available in any of the engineering laboratories

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The Practical Test



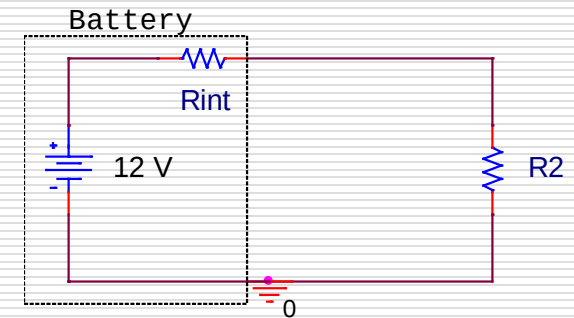
- ❑ At the start of the practical session – 30 minute test.
- ❑ Each student completes test individually
- ❑ Questions will test content mentioned during today's lecture, these slides, the tutorial, the pre-prac assignment and your circuits knowledge up to now

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Example Question



- Consider the figure below
 - The battery has **internal resistance** of 5Ω
 - R_2 is the **load resistance** (eg 2Ω)
 - R_2 must be varied sequentially from $1 - 10\Omega$
- Complete a simulation of the circuit for $R_2 = 5\Omega$
 - Determine the **current** through R_2
 - Determine the **voltage** across R_2
- Sequentially set the value of R_2 to five values in the range $1 - 10\Omega$
 - Determine the **load voltage** (V_2) across R_2 and the **current** for each resistor value



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Practical



- After the test, you will work in groups of two to complete the practical

PRINT OUT PRACTICAL 1 GUIDELINES (found on ClickUP) AND BRING IT TO THE PRACTICAL SESSION!!!!

Use the guidelines to complete the practical and then after completion – log on to ClickUP, start Practical 1 Online Evaluation and fill in your answers

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Practical Assessment - Online



□ Practical assessment breakdown

- Pre-Practicum Test: 50 %
- Practical: 50 %

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Conclusion - Enjoy



□ Contact details for Questions or Feedback

- *Email:* dieter.barnard@up.ac.za
 - **NB:** For all emails use the subject “[EBN 111] *subject*”
- *Office no.:* Eng 1, 13-26
- *Email to make appointment*

□ During practical

- There will be two Demis to assist you during the whole duration of the practical
- Ask questions and remember you can also refer to the practical guide or materials available on the module website